

Problem Set for 26–May and 29–May

Exercise 0 Show that $U \otimes V \simeq V \otimes U$ with the following steps.

1. Construct $\varphi : U \otimes V \rightarrow V \otimes U$ via a 1:1 correspondence (refer to the preceding assignment). Describe φ by tracing the image of simple tensors.
2. Define $\psi : V \otimes U \rightarrow U \otimes V$ analogously.
3. Establish via 1:1 correspondence that both $\psi \circ \varphi$ and $\varphi \circ \psi$ are identical mappings.

取双线性型

$$\Phi : U \otimes V \rightarrow V \otimes U, \quad (u, v) \mapsto v \otimes u.$$

这给出唯一的 $\varphi : U \otimes V \rightarrow V \otimes U$, 简单张量 $u \otimes v$ 的像是 $v \otimes u$.

对称地定义 $\psi : V \otimes U \rightarrow U \otimes V$, 简单张量 $v \otimes u$ 的像是 $u \otimes v$.

由 $\varphi \circ \psi$ 与 $\psi \circ \varphi$ 在简单张量表现为恒等映射, 从而两者就是张量空间的恒等映射. 这说明 φ 与 ψ 是互逆的.

Exercise 1 Show that the following is an isomorphism:

$$\text{Hom}(U, V^*) \simeq \text{Hom}(V, U^*), \quad [u \mapsto f_u] \mapsto [v \mapsto [u \mapsto f_u(v)]].$$

$\text{Hom}(U, \text{Hom}(V, \mathbb{F}))$ 一一对应双线性映射 $U \otimes V \rightarrow \mathbb{F}$, 另一侧亦然.

这题和张量没关系.

Exercise 2 Let V be a finite-dimensional vector space with basis $\{e_i\}$. Let $\{f_i\}$ denote the dual basis of V^* . Define the following mappings:

- $\Delta : \mathbb{F} \rightarrow V \otimes V^*, \quad 1 \mapsto \sum_{i=1}^n e_i \otimes f_i;$
- $\nabla : V \otimes V^* \rightarrow \mathbb{F}, \quad \sum u_i \otimes \varphi_i \mapsto \sum \varphi_i(u_i).$

For $f, g : V \rightarrow V$, determine the composition

$$\mathbb{F} \xrightarrow{\Delta} V \otimes V^* \xrightarrow{(f \otimes g^*)} V \otimes V^* \xrightarrow{\nabla} \mathbb{F}.$$

直接计算, $1 \mapsto \sum e_i \otimes f_i \mapsto f(e_i) \otimes (f_i \circ g) \mapsto \sum f_i g f e_i$, 也就是 $\text{tr}(g \circ f)$.

Exercise 3 Demonstrate that, for finite-dimensional vector spaces, there exists an isomorphism determined via simple tensors:

$$\Phi : V_1^* \otimes V_2^* \otimes V_3 \xrightarrow{\sim} \text{Hom}(V_1 \otimes V_2, V_3), \quad f \otimes g \otimes x \mapsto [a \otimes b \mapsto f(a)g(b)x].$$

Now take $\otimes = \otimes_{\mathbb{R}}$ and $V_i = \mathbb{C}$ (the two-dimensional vector space with basis $\{1, i\}$). We know that the usual multiplication defines a map in $\text{Hom}_{\mathbb{R}}(\mathbb{C} \otimes \mathbb{C}, \mathbb{C})$:

$$\times : \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \rightarrow \mathbb{C}, \quad w \otimes z \mapsto w \cdot z.$$

What is $\Phi^{-1}(\times)$?

记 (e, f) 是 $(1, i)$ 的对偶基. 乘法运算在 $\mathbb{C} \otimes \mathbb{C}$ 的一组基底上的表现是

$$(1 \otimes 1, 1 \otimes i, i \otimes 1, i \otimes i) \mapsto (1, i, i, -1).$$

这组基的对偶基是 $(e \otimes e, e \otimes f, f \otimes e, f \otimes f)$, 从而 $\Phi^{-1}(\times)$ 是张量

$$e \otimes e \otimes 1 + e \otimes f \otimes i + f \otimes e \otimes i - f \otimes f \otimes 1.$$

Exercise 4 (optional) Show that $\Phi^{-1}(\times)$ comprises a sum of no fewer than three simple tensors. Consequently, any multiplication algorithm for complex numbers necessitates at least three real multiplications.

考虑实线性空间对偶 $\mathbb{C}^* \simeq \mathbb{C}$, 只需证明以下 $\mathbb{C} \otimes \mathbb{C} \otimes \mathbb{C}$ 中的张量最少只能写作三个张量的有限和:

$$1 \otimes 1 \otimes 1 + 1 \otimes i \otimes i + i \otimes 1 \otimes i - i \otimes i \otimes 1.$$

说明三个可以:

$$\frac{1}{2}(1+i) \otimes (1+i) \otimes (1+i) + \frac{1}{2}(1-i) \otimes (1-i) \otimes (1-i) - 2(1 \otimes 1 \otimes 1).$$

再说明两个不行. 若

$$1 \otimes (1 \otimes 1 + i \otimes i) + i \otimes (1 \otimes i - i \otimes 1).$$

可以写作两个简单张量的和, 例如 $x \otimes y \otimes z + p \otimes q \otimes r$, 则有

$$\text{span}(1 \otimes 1 + i \otimes i, 1 \otimes i - i \otimes 1) \subseteq \text{span}(y \otimes z, q \otimes r).$$

将二阶张量 $U \otimes V$ 写成矩阵形式, 则上式写作

$$\text{span}\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\right) \subseteq \text{span}(u_1 \cdot v_1^T, u_2 \cdot v_2^T).$$

左式中, 非零线性组合必然给出秩为 2 的矩阵, 矛盾.

这证明了复数乘法的 Gauss 算法: 可以仅做三次实数乘法, 但无法仅做两次实数乘法.

Exercise 5 Let $A \in \mathbb{C}^{m \times m}$ and $B \in \mathbb{C}^{n \times n}$ be normal matrices (i.e., $PP^H = P^H P$). Show that $AX = XB$ if and only if $A^H X = XB^H$.

即证明

$$\ker(A \otimes I - I \otimes B^T) = \ker(A^H \otimes I - I \otimes \overline{B}).$$

记 $P = A \otimes I - I \otimes B^T$, 则 $PP^H = P^H P$. 那么

$$\ker P = \ker P^H P = \ker PP^H = \ker P^H.$$

Exercise 6 Let $A, B \in \mathbb{C}^{n \times n}$ be Hermitian positive-definite matrices, i.e., $A = A^H$ and $u^H \cdot A \cdot u \geq 0$ with equality if and only if $u = \mathbf{0}$. Define the matrix C via component-wise multiplication (the stupid multiplication):

$$C = (c_{i,j}), \quad c_{i,j} = a_{i,j} \cdot b_{i,j}.$$

Show that C is also Hermitian positive-definite

M 是 Hermite 正定的, 当且仅当存在 S 使得 $S^H S = M$. 因此

$$A \otimes B = (R^H R) \otimes (S^H S) = (R \otimes S)^H (R \otimes S)$$

也是 Hermite 正定的. 显然 C 是 $A \otimes B$ 的主子式 (用 $((i_1, j_1), (i_2, j_2))$ 表示 $A \otimes B$ 行列分量坐标, 取所有 $i_1 = i_2$ 且 $j_1 = j_2$ 的分量即可), 因此 C Hermite 正定.

Exercise 6 The tensor product space $U \otimes V$ (as defined in class) arises from the Cartesian product $U \times V$ along with a certain quotient construction. Explain why this quotient makes the resulting linear space larger, for instance:

$$\dim(\mathbb{F}^2 \times \mathbb{F}^3) = 5 < 6 = \dim(\mathbb{F}^2 \otimes \mathbb{F}^3).$$

被商的是以 $U \times V$ 这个集合为基的自由线性空间 (通常是无限维的), 并不是 $U \times V$ 这个直和空间.